

SSQ-LR: Sub-FP16 Recurrent State Quantization for Hybrid Reasoners

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Abstract

This is a living paper shell for a preregistered Mac-first experiment. SSQ-LR tests whether recurrent SSM state in hybrid reasoning models can be quantized below FP16 while preserving quality. No positive result or native systems claim is made yet.

1 Current Gate

The branch must pass state-distribution heterogeneity, state-only quantization sensitivity, and cross-model transfer gates before any GPU validation.

2 Mac Artifact Status

A deterministic synthetic S1 packet validates the readout format before real hybrid-model state dumps. It reports late/early max-abs ratio 8.461, std ratio 3.640, and kurtosis ratio 3.141, producing `SYNTHETIC_PASS_REAL_STATE_DUMPS_NEXT`. This is synthetic-only: it is not model evidence, not a quality result, and not GPU evidence.

3 Next Evidence

The next admissible row is a real SSM-state dump from the smallest available hybrid reasoner. Promotion requires S1 heterogeneity on real states before any state-quantization recipe is tuned.